

Dr. Alex Broughton

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Education	University of California, Irvine	
	<i>Ph.D.</i> , Physics	2024
	University of California, Davis	
	<i>B.S.</i> , Physics	2019
Research Experience	Postdoctoral Research Associate	SLAC National Lab/Stanford University
	2024-present	
	I work on the Legacy Survey of Space and Time (LSST) at the Vera C. Rubin Observatory, within Kavli Institute for Particle Astrophysics and Cosmology (KIPAC) at SLAC/Stanford University. I'm responsible for development, commissioning, and science validation and verification for the Instrument Signature Removal (ISR) for the Data Management Science Pipeline. I'm also involved in the ComCam on-sky calibration plan as part of the TAXICAB calibration team. Additionally, I contribute forward models of image systematics to cosmology in the Dark Energy Science Collaboration (DESC)'s Pixel-to-Object (PO) working group.	
	Graduate Student Researcher	SLAC National Lab/Stanford University
	2022-2023	
	Studied sensor artifacts and impacts on dark energy and dark matter science for ongoing Legacy Survey of Space and Time (LSST) camera and Auxillary Telescope (AuxTel) science and commissioning tasks.	
	Graduate Student Researcher	UC Irvine
	2019-2024	
	Operated LSSTCam and developed computational tools, data processing pipelines, and simulations for star/galaxy characterization, wavefront sensing and guiding, sensor spacing and other electro-optical sensor systematics.	
	Graduate Student Researcher	UC Irvine
	2019-2024	
	Utilized a deep learning approach to improve systematic errors in gamma-ray component models used in testing theories of self-interacting dark matter for the Fermi Large Area Telescope (Fermi-LAT).	
	Research Assistant	UC Davis
	2017-2019	
	Developed Monte Carlo simulations of gamma-ray energy deposits in a liquid Xenon scintillator for the Davis Xenon Experiment (DAX), a test-bed direct-detection dark matter experiment, and the Large Underground Xenon (LUX) experiment.	
	Research Assistant	UC Davis
	2017-2019	
	Conducted simulations of WIMP dark matter energy depositions in a liquid xenon time-projection chamber. Also ran computation and hardware setup for critical high-voltage conduit testing for the LUX-Zepplin (LZ) project.	

Summer Research Assistant
2016

Lawrence Berkeley National Lab (LBNL)

Constructed a demonstration of a smart nanogrid for efficient energy allocation across multiple linked devices. Developed transactive networks by modeling the TROPEC testbed with the Volttron server, using the Local Price-Based Power Distribution Model (LPDM). Building Technology & Urban System Division.

Publications

- Broughton, Alex**, Yousuke Utsumi, Andrés A. Plazas Malagón, Christopher Waters, Craig Lage, Adam Snyder, Andrew Rasmussen, Stuart Marshall, Jim Chiang, Simona Murgia, and Aaron Roodman. “Mitigation of the Brighter-fatter Effect in the LSST Camera”. In: *Publications of the Astronomical Society of the Pacific* 136.4 (Apr. 2024), p. 045003. DOI: 10.1088/1538-3873/ad3aa2. URL: <https://dx.doi.org/10.1088/1538-3873/ad3aa2>.
- Plazas Malagón, Andrés A., Chris Waters, **Alex Broughton**, Eli Rykoff, Agnès Ferté, Merlin Fisher-Levine, and Robert Lupton. “Instrument Signature Removal and Calibration Products for the Rubin Legacy Survey of Space and Time”. In: *arXiv e-prints*, arXiv:2404.14516 (Apr. 2024), arXiv:2404.14516. DOI: 10.48550/arXiv.2404.14516. arXiv: 2404.14516 [astro-ph.IM].
- Roodman, A., A. Rasmussen, A. Bradshaw, E. Charles, J. Chiang, S. W. Digel, R. Dubois, A. S. Johnson, S. Kahn, S. Liang, S. Marshall, H. Neal, A. A. Plazas, K. Reil, E. Rykoff, R. Schindler, T. Schutt, Y. Utsumi, T. Bogart, T. Bond, B. Bowdish, S. Cisneros, A. Eisner, M. Freytag, D. Hascall, T. Lange, J. C. Lazarte, M. Lopez, C. Mendez, S. Newbry, M. Nordby, D. Onoprienko, S. Osier, H. Pollek, B. Qiu, O. Saxton, S. Tether, G. Thayer, M. Turri, J. Banovetz, P. O’Connor, V. Riot, J. Wolfe, C. Lage, D. Polin, A. Snyder, A. Tyson, R. Nichols, S. Ritz, A. Shestakov, D. Wood, **A. Broughton**, H. Park, J. Esteves, A. Barrau, J. Bregeon, C. Combet, G. Dargaud, E. Lagorio, M. Migliore, F. Vezzu, P. Antilogus, P. Astier, G. Daubard, C. Juramy, D. Laporte, T. Guillemin, E. Aubourg, A. Boucaud, C. Parisel, F. Virieux, P. Breugnon, P. Karst, A. Marini, M. Fisher-Levine, and C. Waters. “LSST camera verification testing and characterization”. In: *Ground-based and Airborne Instrumentation for Astronomy X*. Ed. by Julia J. Bryant, Kentaro Motohara, and Joël. R. D. Vernet. Vol. 13096. Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series. July 2024, 130961S, 130961S. DOI: 10.1117/12.3019698.
- Utsumi, Yousuke, Pierre Antilogus, Pierre Astier, John Banovetz, Andrew K. Bradshaw, Johan Bregeon, **Alex Broughton**, Jim Chiang, Celine Combet, Guillaume Dargaud, Seth W. Digel, Johnny Esteves, Thibault Guillemin, Tony Johnson, Claire Juramy, Craig Lage, Shuang Liang, Stuart Marshall, Myriam Migliore, Homer Neal, Renee Nichols, Daniel Polin, Andrew P. Rasmussen, Steve Ritz, Eli Rykoff, Aaron Roodman, Theo Schutt, Adrian Sheshtakov, Adam Snyder, Gregg Thayer, Max Turri, Tony Tyson, and Duncan Wood. “LSST Camera focal plane optimization”. In: *X-Ray, Optical, and Infrared Detectors for Astronomy XI*. Ed. by Andrew D. Holland and Kyriaki Minoglou. Vol. 13103. Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series. Aug. 2024, 131030W, 131030W. DOI: 10.1117/12.3019117.
- Esteves, Johnny H., Yousuke Utsumi, Adam Snyder, Theo Schutt, Alex Broughton, Bahrudin Trbalic, Sidney Mau, Andrew Rasmussen, Andrés A. Plazas Malagón, Andrew Bradshaw, Stuart Marshall, Seth Digel, James Chiang, Eli Rykoff, Chris Waters, Marcelle Soares-Santos, and Aaron Roodman. “Photometry, Centroid and Point-spread Function Measurements in the LSST Camera Focal Plane Using Artificial Stars”. In: *PASP* 135.1053 (Dec. 2023), p. 115003. DOI: 10.1088/1538-3873/ad0a73. URL: <https://dx.doi.org/10.1088/1538-3873/ad0a73>.
- Karwin, Christopher M., **Broughton, Alex**, Simona Murgia, Alexander Shmakov, Mohammadamin Tavakoli, and Pierre Baldi. “Improved modeling of the discrete

component of the Galactic interstellar γ -ray emission and implications for the Fermi-LAT Galactic Center excess”. In: *Phys. Rev. D* 107 (12 June 2023), p. 123032. DOI: 10.1103/PhysRevD.107.123032. URL: <https://link.aps.org/doi/10.1103/PhysRevD.107.123032>.

Shmakov, Alexander, Mohammadamin Tavakoli, Pierre Baldi, Christopher M. Karwin, **Alex Broughton**, and Simona Murgia. “Deep learning models of the discrete component of the Galactic interstellar γ -ray emission”. In: *Phys. Rev. D* 107 (6 Mar. 2023), p. 063018. DOI: 10.1103/PhysRevD.107.063018. URL: <https://link.aps.org/doi/10.1103/PhysRevD.107.063018>.

Brown, Richard, Stephen Czarnecki, Michael Spears, Bruce Nordman, **Alex Broughton**, Michael Von Hippel, and Anna Liao. *Price Based Local Power Distribution Management System (Local Power Distribution Manager) v1.0*. [Computer Software] <https://doi.org/10.11578/dc.20210416.76>. Nov. 2016. DOI: 10.11578/dc.20210416.76. URL: <https://doi.org/10.11578/dc.20210416.76>.

Honors & Awards

Department of Energy Office of Science Graduate Student Research (SCGSR) Fellowship, Dept. of Energy (DOE), 2022
Data Science Fellowship (DSF) 2021, Physical Sciences Machine Learning (PSML) NEXUS, UC Irvine, 2021

Conference Presentations & Invited Talks

Broughton, Alex. “Constraining Astrophysical Parameters with CCD Systematics,” Image Sensors for Precision Astronomy (ISPA), SLAC National Lab, March 2024.
Broughton, Alex. “Instrument-to-Science: Enabling the Legacy Survey of Space and Time (LSST),” AstroLunch Seminar Series, University of Washington (UW), Jan. 2024.
Broughton, Alex. “Pixels-to-Science: Enabling the Legacy Survey of Space and Time,” Rubin@SLAC Seminar Series, SLAC National Lab/Stanford University, Jan. 2024.
Broughton, Alex. “Instrument-to-Science: Enabling the Legacy Survey of Space and Time (LSST),” Colloquium, CalPoly SLO, 2023.
Broughton, Alex. “The Brighter-Fatter Effect in LSSTCam: Correction and Influence on Science,” KIPAC Rubin Science Talk, Stanford University, 2023.
Broughton, Alex. “Impact of the BFE on Cosmic Shear,” Dark Energy Science Collaboration (DESC) Meeting, SLAC National Lab, 2023.
Broughton, Alex. “Modeling γ -Ray Emission with Deep Learning and Implications for Dark Matter,” PSML Nexus Seminar Series, Irvine, CA 2021.
Broughton, Alex. “Bias Persistence in Run 4 LSST Camera Testing,” Dark Energy Science Collaboration (DESC) Meeting, Virtual, 2021.
Broughton, Alex. “Characterizing Newly Discovered ITL Sensor ‘Comb’ Pattern,” Dark Energy Science Collaboration Meeting, Virtual, 2020.
Broughton, Alex. “Simulation of Light Yield and Collection Efficiency in a Liquid Xenon Detector,” UC Davis Undergraduate Research Conference, Davis, CA, 2018.

Outreach & Leadership

Rubin 2036 - SOC/LOC Organizer
2026

Workshop dedicated to planning future survey and instrumentation for the Simonyi Telescope post-LSST’s 10-year survey.

La Serena Rubin Science Workshop - Lead Organizer
2025

Workshop dedicated to early Rubin Science and synergies with other NSF/NOIRLab/AURA telescopes, bring in project team members and local Universidad de La Serena colleagues.

**Co-founder KIPAC-Chile
Engagement Program**

2024-present

The KIPAC-Chile Engagement Program, launched through a KIPAC Innovation Grant, gives KIPAC members who travel to Chile access to specialized funds meant to help develop professional and personal connections, and spark new research collaborations with scientists at institutions across Chile. Managed a budget of \$15,000 and organized the travel of 10 institutional researcher to give talks at all universities in Chile with PhD programs in astronomy/physics. Developed new metrics to track engagement and impact after 2 years.

Save The Bay (Non-Profit)

2023-present

Permanent active member for political advocacy, education, restoration, and conservation of native wildlife and habitat in the San Francisco Bay Area.

**Teaching/
Mentorship
Experience**

Teaching Assistant, PHYSICS 2: Introduction to Mathematical Methods for Physics,
UC Irvine, Fall 2019.

Physics Tutor, UC Irvine, 2019.

Physics Tutor, UC Davis Physics Club, UC Davis, 2017-2019.